DDD-P-52
DECEMBER 22, 1959
(See 6.3)

FEDERAL SPECIFICATION

PAD, CLOTH, COPPER-SULFATE IMPREGNATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers copper-sulfate impregnated pads for first-aid use against white phosphorus on the skin.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC–T–191—Textile Test Methods.
PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking For Domestic Shipments (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL—B—10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

521493/59/204

FSC 6510

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American Pharmaceutical Association Publication:

National Formulary.

(Application for copies should be addressed to the J. B. Lippincott Company, Philadelphia, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Cloth.*—The cloth shall be made of thoroughly clean cotton, free from waste, and shall be of the following construction.

3.1.1.1 *Weave.*—The cloth shall consist of warp and filling ground yarns and warp pile yarns. The cloth shall be woven with a terry weave. The thread count of the woven fabric shall be 68 threads per inch in the warp (combined ground and pile thread count) and 32 threads per inch in the filling (plus or minus tolerance of 4 threads in the warp and 3 threads in the filling, but not more than plus or minus tolerance of 5 threads per square inch), when tested as specified in 4.4.1.1.

3.1.1.2 *Yarn.*—The ground and pile warps shall be either two-ply or single-ply yarn of equivalent size; the filling shall be single-ply yarn.

3.1.1.3 *Finish.*—The cloth shall be fully bleached and shall be free from sizing.

3.1.1.4 *Weight.* — The terry cloth shall weigh a minimum of 9 ounces per square yard when tested in accordance with 4.4.1.2.

3.1.1.5 *Breaking strength.*—The cloth shall have a minimum breaking strength of 35 pounds in the warp and filling when tested in accordance with 4.4.1.3.

3.1.2 *Copper sulfate* ($Cu\ SO_4 \cdot 5H_2O$).— The copper sulfate shall be in accordance with the requirements of the current edition of the National Formulary.

3.2 Completed pad. — The completed pad shall consist of a piece of terry cloth as described herein, measuring 3 by 3 inches square (plus or minus $\frac{1}{16}$ inch in either direction), and shall contain not less than 1.0 grams nor more than 1.2 grams of copper sulfate when tested as described in 4.4.2.2. The cloth pad shall be completely and uniformly impregnated with copper sulfate.

3.3 Directions for use.—Each copper sulfate impregnated pad shall be furnished with directions for use (see 5.3.1 and 5.3.2).

Directions for Use.

For use as first-aid against white phosphorus (WP) on skin.

1. Douse burn with canteen water. Wet pad with canteen water.
2. Press wet pad on the white-phosphorus particles. Then squeeze the liquid on the burned area.
3. Carefully pick out large black particles of phosphorus from skin with knife, bayonet, or match stick.
4. Wipe off any remaining particles with wet pad.
5. Discard pad. Do not use pad as dressing for burn.

3.4 Workmanship.—Workmanship shall be first class throughout. The copper-sulfate impregnated pads shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without

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the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.** — Sampling shall be conducted in accordance with Military Standard MIL–STD–105. The acceptance number shall be zero (0) for all sample sizes.

	Inspection level	Minimum sample size
Dimensional examination	L2	3 pkgs. of 3s.
Chemical testing	L2	1 pkg. of 3s.

4.2.1 *Components.*

4.2.1.1 *Copper sulfate.*—One ounce of each lot of copper sulfate used in the construction of the copper-sulfate impregnated pads shall be selected for chemical analysis (see 4.1.2 and 4.4.2.1).

4.2.1.2 *Terry cloth.*—Prior to being impregnated with copper sulfate, one square yard of terry cloth used in the construction of the copper-sulfate impregnated pads shall be selected for tests (see 4.4.1.1, 4.4.1.2, and 4.4.1.3).

4.2.2 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.**—The copper-sulfate impregnated pads shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 *Dimensional examination.*—The copper-sulfate impregnated pads shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. — Tests shall be conducted, as necessary, to determine compliance with all requirements contained in this specification.

4.4.1 *Terry cloth.*

4.4.1.1 *Yarns per inch.*—The yarns per inch shall be determined in accordance with method 5050 of Federal Specification CCC–T–191.

4.4.1.2 *Weight of cloth.* — The weight of cloth shall be determined in accordance with method 5041 of Federal Specification CCC–T–191.

4.4.1.3 *Breaking strength.*—The breaking strength of the cloth shall be determined in accordance with method 5102 of Federal Specification CCC–T–191.

4.4.2 *Copper sulfate* ($Cu\ SO_4 \cdot 5H_2O$).

4.4.2.1 *Copper sulfate.*—The copper sulfate shall be in accordance with the requirements listed for "Cupric Sulfate" in the current edition of the National Formulary.

4.4.2.2 *Copper sulfate in pad.*—Place one copper-sulfate pad in a beaker, add 75 ml. of distilled water, 5 ml. of acetic acid and boil for two minutes. Allow to cool and add 3.5 g. of potassium iodide. Titrate 20 ml. of 0.1*N* sodium thiosulfate into the solution, add starch test solution, and continue titration of liberated iodine to the end point. Each ml. of 0.1*N* sodium thiosulfate is equivalent to 24.97 mg. of $Cu\ SO_4 \cdot 5H_2O$.

Note.—At times, after the end point is reached, if the solution is allowed to stand a blue color will return. This is to be disregarded, and no further titration should take place.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

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5.1 Packaging.

5.1.1 *Unit of issue.*—Three pads, as specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Unit package	*Intermediate package*	*Exterior container*
1 unit	50 units	800 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Individual package.*—Each pad shall be individually packaged in an unsealed glassine paper, cellulose acetate, or polyethylene envelope of suitable size and design.

5.1.3.2 *Unit package.*—Each unit (3 pads) shall be packaged in a bag measuring approximately 4¼ by 5½ inches fabricated from virgin polyethylene seamless tubing having an average thickness of not less than 0.004 inch, with no wall thickness at any point below 0.00325 inch. Excess air shall be expelled or evacuated, and closure shall be effected by heat sealing.

5.1.3.3 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specifications PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specifications PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I, or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking (nonmilitary).

5.3.1 *Unit packages.* — Unless otherwise specified, packages shall be marked with the name and quantity contained therein, and the names of the contractor and manufacturer.

5.3.2 *Exterior containers.*—In addition to any special markings in the contract or order, exterior shipping containers shall be marked in accordance with Federal Standard No. 123.

5.4 Marking (Military agencies).

5.4.1 *Individual package.*—Each individual envelope shall have printed on one face the required directions for use (see 3.3).

5.4.2 *Unit package.*—Each unit bag shall have printed on both faces the stock number, item identification, unit, quantity, lot or control number, name of contractor, name of manufacturer when manufacturer is not the contractor, and the required directions for use (see 3.3).

5.4.3 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129. Lot (control) number shall be shown.

5.4.4 *Exterior container.* — Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), 6.5, and 7.5.1 shall not be applicable. Lot (control) number shall be shown. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Supersession data.—This specification includes the requirements of Military Specification MIL–P–15355A, dated January 26, 1953.

6.4 Transportation description.—Transportation description applicable to this item is:

Dressings, Surgeons'.

Carload minimum weight 24,000 pounds subject to Rule 34 of the Uniform Freight Classification.

Truckload minimum weight 24,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Preparing activity:
Navy—Bureau of Medicine and Surgery (MMSA)

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producers' own drawings. description and contents for the